

```

1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70      /* CMP-135 SDTM MH Creation
71      Domain: MH*/;
72
73      options validvarname=upcase nofmterr dlcreatedir;
74
75      libname raw "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
NOTE: Libref RAW was successfully assigned as follows:
      Engine:          V9
      Physical Name:   /home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1
76      libname sdtm "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";
NOTE: Libref SDTM was successfully assigned as follows:
      Engine:          V9
      Physical Name:   /home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm
77
78      libname valxpt xport
79      "/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/mh.xpt";
NOTE: Libref VALXPT was successfully assigned as follows:
      Engine:          XPORT
      Physical Name:   /home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/mh.xpt
80
81
82      %macro make_date(invar=, outdt=);
83          if vtype(&invar) = "N" then do;
84              if not missing(&invar) then do;
85                  if &invar > 100000 then &outdt = datepart(&invar);
86                  else &outdt = &invar;
87              end;
88          end;
89          else do;
90              _dtm = input(strip(&invar), anydtdtm.);
91              &outdt = datepart(_dtm);
92
93              if missing(&outdt) then
94                  &outdt = input(strip(&invar), anydtdte.);
95          end;
96      %mend;
97
98
99
100     data ex_prep;
101         set raw.ex;
102
103         %make_date(invar=EXSTDN, outdt=exdt);
104
105         format exdt date9.;
106
107         keep SUBJECT exdt;
108     run;

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).
      103:1    103:1
NOTE: There were 1904 observations read from the data set RAW.EX.
NOTE: The data set WORK.EX_PREP has 1904 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds
      memory              1663.43k
      OS Memory           25000.00k
      Timestamp           06/29/2026 07:32:47 PM
      Step Count          519   Switch Count   2
      Page Faults         0
      Page Reclaims      211
      Page Swaps          0
      Voluntary Context Switches 19
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 264

109
110     proc sql;
111         create table ex_first as
112         select
113             SUBJECT,

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114         min(exdt) as rfstdt format=date9.
115     from ex_prep
116     where not missing(exdt)
117     group by SUBJECT;
NOTE: Table WORK.EX_FIRST created, with 24 rows and 2 columns.

```

```

118     quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              5858.09k
      OS Memory           29608.00k
      Timestamp           06/29/2026 07:32:47 PM
      Step Count          520  Switch Count   5
      Page Faults         0
      Page Reclaims       153
      Page Swaps           0
      Voluntary Context Switches 17
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 280

```

```

119
120
121     data dov_prep;
122         set raw.dov;
123
124         %make_date(invar=VISDTN, outdt=dovdt);
125
126         length INSTANCENAME_U $100 FOLDERNAME_U $100;
127
128         INSTANCENAME_U = upcase(strip(INSTANCENAME));
129         FOLDERNAME_U   = upcase(strip(FOLDERNAME));
130
131         format dovdt date9.;
132
133         keep SUBJECT INSTANCENAME_U FOLDERNAME_U VISDTN dovdt;
134     run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).

124:1 124:1

NOTE: There were 644 observations read from the data set RAW.DOV.

NOTE: The data set WORK.DOV_PREP has 644 observations and 5 variables.

NOTE: DATA statement used (Total process time):

```

      real time           0.00 seconds
      user cpu time       0.00 seconds
      system cpu time     0.00 seconds
      memory              1403.46k
      OS Memory           25000.00k
      Timestamp           06/29/2026 07:32:47 PM
      Step Count          521  Switch Count   2
      Page Faults         0
      Page Reclaims       107
      Page Swaps           0
      Voluntary Context Switches 17
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 528

```

```

135
136
137     proc sql;
138         create table mh_with_dov as
139         select
140             a.*,
141             b.dovdt as mh_collection_dt format=date9.
142         from raw.mh as a
143         left join dov_prep as b
144             on a.SUBJECT = b.SUBJECT
145             and upcase(strip(a.INSTANCENAME)) = b.INSTANCENAME_U;
NOTE: Table WORK.MH_WITH_DOV created, with 200 rows and 46 columns.

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146     quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      user cpu time       0.01 seconds
      system cpu time     0.00 seconds

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```

memory                26628.82k
OS Memory              49388.00k
Timestamp              06/29/2026 07:32:47 PM
Step Count             522   Switch Count  2
Page Faults            0
Page Reclaims          499
Page Swaps             0
Voluntary Context Switches 29
Involuntary Context Switches 1
Block Input Operations 0
Block Output Operations 792

```

```

147
148
149      proc sql;
150          create table dov_screen as
151          select
152              SUBJECT,
153              min(dovdt) as screening_dt format=date9.
154          from dov_prep
155          where FOLDERNAME_U = "SCREENING"
156             or INSTANCENAME_U = "SCREENING"
157             or index(FOLDERNAME_U, "SCREEN") > 0
158             or index(INSTANCENAME_U, "SCREEN") > 0
159          group by SUBJECT;
NOTE: Table WORK.DOV_SCREEN created, with 24 rows and 2 columns.

```

```

160      quit;
NOTE: PROCEDURE SQL used (Total process time):
real time              0.00 seconds
user cpu time          0.00 seconds
system cpu time        0.00 seconds
memory                6017.34k
OS Memory              30888.00k
Timestamp              06/29/2026 07:32:47 PM
Step Count             523   Switch Count  2
Page Faults            0
Page Reclaims          150
Page Swaps             0
Voluntary Context Switches 10
Involuntary Context Switches 1
Block Input Operations 0
Block Output Operations 280

```

```

161
162
163      proc sql;
164          create table mh_merge as
165          select
166              a.*,
167              b.screening_dt,
168              c.rfstdt
169          from mh_with_dov as a
170          left join dov_screen as b
171              on a.SUBJECT = b.SUBJECT
172          left join ex_first as c
173              on a.SUBJECT = c.SUBJECT;
NOTE: Table WORK.MH_MERGE created, with 200 rows and 48 columns.

```

```

174      quit;
NOTE: PROCEDURE SQL used (Total process time):
real time              0.00 seconds
user cpu time          0.00 seconds
system cpu time        0.01 seconds
memory                6734.46k
OS Memory              31536.00k
Timestamp              06/29/2026 07:32:47 PM
Step Count             524   Switch Count  2
Page Faults            0
Page Reclaims          207
Page Swaps             0
Voluntary Context Switches 12
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 776

```

175

```

176
177
178     data mh_pre;
179         set mh_merge;
180
181         length
182             STUDYID  $6
183             DOMAIN   $2
184             MHTERM    $64
185             MHLLT     $35
186
187 WARNING: Length of character variable MHTERM has already been set.
188 Use the LENGTH statement as the very first statement in the DATA STEP to declare the length of a character variable.
189
190         MHDECOD $40
191         MHHLT   $65
192         MHHLGT  $86
193         MHCAT   $23
194         MHBODSYS $67
195         MHSOC   $67
196         MHDTC   $10
197         MHSTDTC $10
198         MHENDTC $10
199         MHENRTPT $7
200         MHENTPT  $10
201         USUBJID  $40;
202
203     STUDYID = "CMP135";
204     DOMAIN  = "MH";
205     USUBJID = catx("-", STUDYID, strip(SUBJECT));
206
207     MHTERM = strip(MHTERM);
208     MHLLT  = strip(MHTERM_LL);
209     MHDECOD = strip(MHTERM_PT);
210     MHHLT   = strip(MHTERM_HLT);
211     MHHLGT  = strip(MHTERM_HLGT);
212
213     MHLLTCD_N = input(strip(vvalue(MHTERM_LL_CODE)), best.);
214     MHPTCD_N  = input(strip(vvalue(MHTERM_PT_CODE)), best.);
215     MHHLTCD_N = input(strip(vvalue(MHTERM_HLT_CODE)), best.);
216     MHHLGTCD_N = input(strip(vvalue(MHTERM_HLGT_CODE)), best.);
217     MHBDSYCD_N = input(strip(vvalue(MHTERM_SOC_CODE)), best.);
218     MHSOCCD_N  = input(strip(vvalue(MHTERM_SOC_CODE)), best.);
219
220     MHCAT = "GENERAL MEDICAL HISTORY";
221     MHBODSYS = strip(MHTERM_SOC);
222     MHSOC = strip(MHTERM_SOC);
223
224     if not missing(mh_collection_dt) then mhdt = mh_collection_dt;
225     else mhdt = screening_dt;
226
227     MHDTC = put(mhdt, yymmdd10.);
228
229     if not missing(MHSTDN_YY) then do;
230         if not missing(MHSTDN_MM) and not missing(MHSTDN_DD) then
231             MHSTDTC = cats(put(MHSTDN_YY, z4.), "-", put(MHSTDN_MM, z2.), "-", put(MHSTDN_DD, z2.));
232         else if not missing(MHSTDN_MM) then
233             MHSTDTC = cats(put(MHSTDN_YY, z4.), "-", put(MHSTDN_MM, z2.));
234         else
235             MHSTDTC = cats(put(MHSTDN_YY, z4.));
236     end;
237
238     if not missing(MHENDN_YY) then do;
239         if not missing(MHENDN_MM) and not missing(MHENDN_DD) then
240             MHENDTC = cats(put(MHENDN_YY, z4.), "-", put(MHENDN_MM, z2.), "-", put(MHENDN_DD, z2.));
241         else if not missing(MHENDN_MM) then
242             MHENDTC = cats(put(MHENDN_YY, z4.), "-", put(MHENDN_MM, z2.));
243         else
244             MHENDTC = cats(put(MHENDN_YY, z4.));
245     end;
246
247     if not missing(mhdt) and not missing(rfstdt) then do;
248         if mhdt >= rfstdt then MHDY = mhdt - rfstdt + 1;
249         else MHDY = mhdt - rfstdt;
250     end;
251
252     if MHONG = 1 then do;
253         MHENRTPT = "ONGOING";
254         MHENTPT = put(screening_dt, yymmdd10.);
255     end;
256     else do;
257         MHENRTPT = "";
258     end;

```

```

254      MHENTPT = "";
255  end;
256
257      keep
258          STUDYID DOMAIN MHTERM MHLT MHLTCD_N MHDECOD MHPTCD_N
259          MHLT MHLTCD_N MHLGT MHLGTCD_N MHCAT MHBODSYS MHBDSYCD_N
260          MHSOC MHSOCCD_N MHDTC MHSTDTC MHENDTC MHDY MHENRTPT MHENTPT
261          USUBJID;
262  run;

```

NOTE: Character values have been converted to numeric values at the places given by: (Line):(Column).

248:8

NOTE: There were 200 observations read from the data set WORK.MH_MERGE.

NOTE: The data set WORK.MH_PRE has 200 observations and 23 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	1872.31k
OS Memory	26532.00k
Timestamp	06/29/2026 07:32:47 PM
Step Count	525 Switch Count 2
Page Faults	0
Page Reclaims	207
Page Swaps	0
Voluntary Context Switches	9
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	272

```

263
264
265      proc sort data=mh_pre out=mh_sorted;
266          by STUDYID USUBJID MHTERM MHSTDTC MHENDTC;
267  run;

```

NOTE: There were 200 observations read from the data set WORK.MH_PRE.

NOTE: The data set WORK.MH_SORTED has 200 observations and 23 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	1743.53k
OS Memory	26804.00k
Timestamp	06/29/2026 07:32:47 PM
Step Count	526 Switch Count 2
Page Faults	0
Page Reclaims	126
Page Swaps	0
Voluntary Context Switches	11
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	272

```

268
269      data sdtm.mh;
270          retain
271              STUDYID DOMAIN MHSEQ MHTERM MHLT MHLTCD MHDECOD MHPTCD
272              MHLT MHLTCD MHLGT MHLGTCD MHCAT MHBODSYS MHBDSYCD
273              MHSOC MHSOCCD MHDTC MHSTDTC MHENDTC MHDY MHENRTPT MHENTPT
274              USUBJID;
275
276          length
277              STUDYID $6
278              DOMAIN $2
279              MHTERM $64
280              MHLT $35
281              MHDECOD $40
282              MHLT $65
283              MHLGT $86
284              MHCAT $23
285              MHBODSYS $67
286              MHSOC $67
287              MHDTC $10
288              MHSTDTC $10
289              MHENDTC $10
290              MHENRTPT $7
291              MHENTPT $10

```

```

292         USUBJID  $40;
293
294         set mh_sorted;
295         by STUDYID USUBJID;
296
297         if first.USUBJID then MHSEQ = 1;
298         else MHSEQ + 1;
299
300         MHLTCD   = MHLTCD_N;
301         MHPTCD   = MHPTCD_N;
302         MHLTCD   = MHLTCD_N;
303         MHLGTC   = MHLGTC_N;
304         MHBDSYCD = MHBDSYCD_N;
305         MHSOCCD  = MHSOCCD_N;
306
307         label
308             STUDYID = "Study Identifier"
309             DOMAIN  = "Domain Abbreviation"
310             MHSEQ    = "Sequence Number"
311             MHTERM   = "Reported Term for the Medical History"
312             MHLT     = "Lowest Level Term"
313             MHLTCD   = "Lowest Level Term Code"
314             MHDECOD  = "Dictionary-Derived Term"
315             MHPTCD   = "Preferred Term Code"
316             MHLT     = "High Level Term"
317             MHLTCD   = "High Level Term Code"
318             MHLGT    = "High Level Group Term"
319             MHLGTC   = "High Level Group Term Code"
320             MHCAT    = "Category for Medical History"
321             MHBODSYS = "Body System or Organ Class"
322             MHBDSYCD = "Body System or Organ Class Code"
323             MHSOC    = "Primary System Organ Class"
324             MHSOCCD  = "Primary System Organ Class Code"
325             MHDTC    = "Date/Time of History Collection"
326             MHSTDTC  = "Start Date/Time of Medical History Event"
327             MHENDTC  = "End Date/Time of Medical History Event"
328             MHDY     = "Study Day of History Collection"
329             MHENRTPT = "End Relative to Reference Time Point"
330             MHENTPT  = "End Reference Time Point";
331     /*         USUBJID  = "Unique Subject Identifier"; */
332
333     keep
334         STUDYID DOMAIN MHSEQ MHTERM MHLT MHLTCD MHDECOD MHPTCD
335         MHLT MHLTCD MHLGT MHLGTC MHCAT MHBODSYS MHBDSYCD
336         MHSOC MHSOCCD MHDTC MHSTDTC MHENDTC MHDY MHENRTPT MHENTPT
337         USUBJID;
338 run;

```

WARNING: Multiple lengths were specified for the variable MHTERM by input data set(s). This can cause truncation of data.

NOTE: There were 200 observations read from the data set WORK.MH_SORTED.

NOTE: The data set SDTM.MH has 200 observations and 24 variables.

NOTE: DATA statement used (Total process time):

real time	0.01 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	1229.90k
OS Memory	26024.00k
Timestamp	06/29/2026 07:32:47 PM
Step Count	527
Page Faults	0
Page Reclaims	91
Page Swaps	0
Voluntary Context Switches	33
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	272

```

339
340
341     proc sort data=sdm.mh out=mh_created;
342         by STUDYID USUBJID MHSEQ;
343     run;

```

NOTE: There were 200 observations read from the data set SDTM.MH.

NOTE: The data set WORK.MH_CREATED has 200 observations and 24 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.01 seconds

```

memory          1755.71k
OS Memory       26804.00k
Timestamp       06/29/2026 07:32:47 PM
Step Count      528   Switch Count  2
Page Faults     0
Page Reclaims   121
Page Swaps      0
Voluntary Context Switches  22
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations  280

```

```

344
345     proc sort data=valxpt.mh out=mh_validate;
346         by STUDYID USUBJID MHSEQ;
347     run;

```

NOTE: There were 200 observations read from the data set VALXPT.MH.
 NOTE: The data set WORK.MH_VALIDATE has 200 observations and 24 variables.
 NOTE: PROCEDURE SORT used (Total process time):

```

real time       0.00 seconds
user cpu time    0.00 seconds
system cpu time  0.00 seconds
memory          24336.01k
OS Memory       48316.00k
Timestamp       06/29/2026 07:32:47 PM
Step Count      529   Switch Count  2
Page Faults     0
Page Reclaims   763
Page Swaps      0
Voluntary Context Switches  31
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations  264

```

```

348
349     ods rtf file="/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/MH_PROC_COMPARE.rtf";
NOTE: Writing RTF Body file: /home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/MH_PROC_COMPARE.rtf
350
351     title "PROC COMPARE: SDTM MH Created vs Validation MH";
352
353     proc compare base=mh_validate compare=mh_created listall criterion=0.0000001;
354         id STUDYID USUBJID MHSEQ;
355     run;

```

NOTE: There were 200 observations read from the data set WORK.MH_VALIDATE.
 NOTE: There were 200 observations read from the data set WORK.MH_CREATED.
 NOTE: PROCEDURE COMPARE used (Total process time):

```

real time       0.01 seconds
user cpu time    0.02 seconds
system cpu time  0.00 seconds
memory          2356.40k
OS Memory       27052.00k
Timestamp       06/29/2026 07:32:47 PM
Step Count      530   Switch Count  0
Page Faults     0
Page Reclaims   372
Page Swaps      0
Voluntary Context Switches  0
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations  8

```

```

356
357     ods rtf close;
358
359
360     title "Created MH";
361     proc print data=mh_created(obs=5);
362     run;

```

NOTE: There were 5 observations read from the data set WORK.MH_CREATED.
 NOTE: PROCEDURE PRINT used (Total process time):

```

real time       0.02 seconds
user cpu time    0.02 seconds
system cpu time  0.00 seconds
memory          1045.90k

```

OS Memory	26276.00k	
Timestamp	06/29/2026 07:32:48 PM	
Step Count	531	Switch Count 0
Page Faults	0	
Page Reclaims	65	
Page Swaps	0	
Voluntary Context Switches	0	
Involuntary Context Switches	1	
Block Input Operations	0	
Block Output Operations	16	

```
363
364     title "Validation MH";
365     proc print data=mh_validate(obs=5);
366     run;
```

NOTE: There were 5 observations read from the data set WORK.MH_VALIDATE.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.02 seconds	
user cpu time	0.02 seconds	
system cpu time	0.01 seconds	
memory	793.03k	
OS Memory	26276.00k	
Timestamp	06/29/2026 07:32:48 PM	
Step Count	532	Switch Count 0
Page Faults	0	
Page Reclaims	61	
Page Swaps	0	
Voluntary Context Switches	0	
Involuntary Context Switches	0	
Block Input Operations	0	
Block Output Operations	16	

```
367
368     title;
369
370     OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
380
```